

FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

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ABSTRACT

Designing a continuous flow process demands coordinated reasoning across fluid dynamics, heat and mass transfer, materials compatibility, and safety. We present FlowPilot, a multi-agent AI system that connects chemical interpretation with deterministic engineering calculation and specialist review. Standardized intake captures the batch protocol, chemist objectives, and laboratory constraints; a structured literature corpus and 2,537 handbook-derived rules support the design process. A seven-agent council evaluates candidate operating conditions, which are reconciled with equipment capabilities before a process diagram and traceable design record are returned. Matched architecture comparisons use three chemistry cases, repeated generation, and fixed-criteria evaluation by independent LLM judges informed by deterministic checks. Across the five-model cohort documented in the ESI, FlowPilot achieves higher mean benchmark scores and fewer critical flags than matched one-shot generation. Internal ablations examine the contribution of council review, while archived model-matrix experiments characterize configuration sensitivity. Two multistage chemistries, photoredox Giese addition followed by aerobic oxidation and DPDTC-mediated amidation, are selected for prospective laboratory evaluation. FlowPilot provides an auditable starting point for experimental development rather than a substitute for chemical judgment or measured validation.

INTRODUCTION

Continuous flow chemistry has matured into one of the main technologies^{1,2} for pharmaceutical³, fine-chemical⁴, and on-demand manufacturing⁵, offering orders-of-magnitude improvements in heat and mass transfer⁶, intrinsically safer handling of hazardous intermediates⁷, access to short residence times², and ready accommodation of photochemistry⁸, electrochemistry⁹, liquid-phase oxidation¹⁰, and multi-step coupling^{11–13}. These features can reduce solvent use, energy demand, and equipment footprint when the process is appropriately designed, supporting the resource-efficiency goals of sustainable manufacturing. A mature commercial ecosystem now supplies virtually everything a flow chemist needs¹⁴, yet adoption

remains gated less by what flow can do than by what it costs to design: the gap between a published batch protocol and a working flow process is typically closed by a small team over weeks of iteration; even within established translation paradigms such as microwave-to-flow, each new chemistry class typically requires re-derivation of operating conditions¹⁵. The bottleneck has shifted from running flow chemistry to designing flow processes.

Designing a flow process is a chain of interlocking decisions across fluid dynamics^{16,17}, mass transfer¹⁸, heat transfer¹⁹, materials compatibility, kinetics, and safety, where each decision constrains the next^{20,21}. This coupled, multi-domain design surface is forbidding for an individual chemist. Existing tools address fragments of it: classical process simulators²² solve the engineering rigorously but require a pre-specified flowsheet, and reinforcement-learning approaches relax this restriction only under simplified process abstractions²³; Bayesian and active-learning optimizers^{24–28} refine a known process but cannot propose one; retrosynthesis tools^{29–31} answer what to make but not how to run it in flow. Handbook knowledge^{20,21,32} encodes the rules of thumb an experienced flow chemist relies on, but this knowledge is dispersed across textbook chapters, qualitative in form, and almost never operationalized computationally. The remaining challenge is to connect a chemistry brief to equipment-aware flow design while making these handbook-level fundamentals available for systematic review.

Recent large-language-model (LLM) copilots for chemistry^{33,34} have expanded what AI can do for a chemist, from synthesis planning to autonomous experimentation^{35–38}, but these efforts remain largely batch-oriented. When a monolithic LLM is prompted directly for a flow reactor design, it must handle chemistry, kinetics, fluid mechanics, heat transfer, and safety within a single inference pass^{39,40}, without a separate calculation and reconciliation workflow. A fluent answer therefore does not by itself establish cross-domain consistency or show which constraints were checked.

The structure of flow process design, a coupled problem solved by a team of complementary specialists, maps naturally onto a multi-agent architecture in which each agent owns one domain and a separate orchestrator integrates their judgments. The organizing principle of this work is therefore: LLMs interpret chemistry; deterministic code, anchored in handbook fundamentals, computes.

We introduce FlowPilot, a multi-agent AI pipeline purpose-built for continuous flow chemistry process design. Given a batch protocol, the system returns a flow-process proposal accompanied by a process diagram, traceable engineering calculations, literature analogies, and explicit uncertainty flags. The workflow couples five operations: (i) standardized intake that records the protocol, objective, historical evidence, inventory, and chemist hypotheses; (ii) chemistry interpretation and plan-aware retrieval from a

curated flow-chemistry corpus; (iii) deterministic engineering calculations supported by handbook-derived rules; (iv) a seven-agent council that reviews and revises candidate designs under bounded domain authority; and (v) inventory reconciliation and rendering from a common final design record. Candidate selection considers the chemist's objective and process constraints. Intensification is evaluated where justified, rather than treated as evidence that any shorter residence time will preserve reaction performance.

This work presents the hybrid FlowPilot architecture and evaluates the contribution of its components. Literature and rule-base analyses establish the knowledge coverage, and a retrieval benchmark tests whether chemistry-aware ranking improves the relevance of selected analogies. Matched one-shot versus full-pipeline comparisons then assess delivered designs using a fixed rubric across local and commercial models. Internal module ablations and an earlier isoxazole cycloaddition model-matrix study⁴¹ provide complementary evidence about council behavior. Finally, photoredox Giese addition followed by aerobic oxidation⁴² and telescoped DPDTC-mediated amidation⁴³ provide two connected-process cases for prospective laboratory evaluation. The main figures follow this progression from architecture and knowledge resources to computational benchmarking and experimental application.

RESULTS

FlowPilot was designed to translate a batch chemistry protocol into a continuous-flow process proposal while keeping chemical interpretation, deterministic engineering calculation, and expert review as separate operations (Figure 1). The input layer accepts the protocol, design objective, and laboratory inventory; standardized follow-up questions resolve missing chemistry, constraints, and relevant experimental history. The upstream module converts the confirmed input into a chemistry plan, including species identification, reaction classification, stream-separation logic, sensitivity flags, and possible flow-specific advantages. Retrieval and engineering calculations then support an initial proposal containing reactor geometry, stream assignments, residence-time assumptions, and operating conditions. These initial values remain subject to council review and equipment reconciliation.

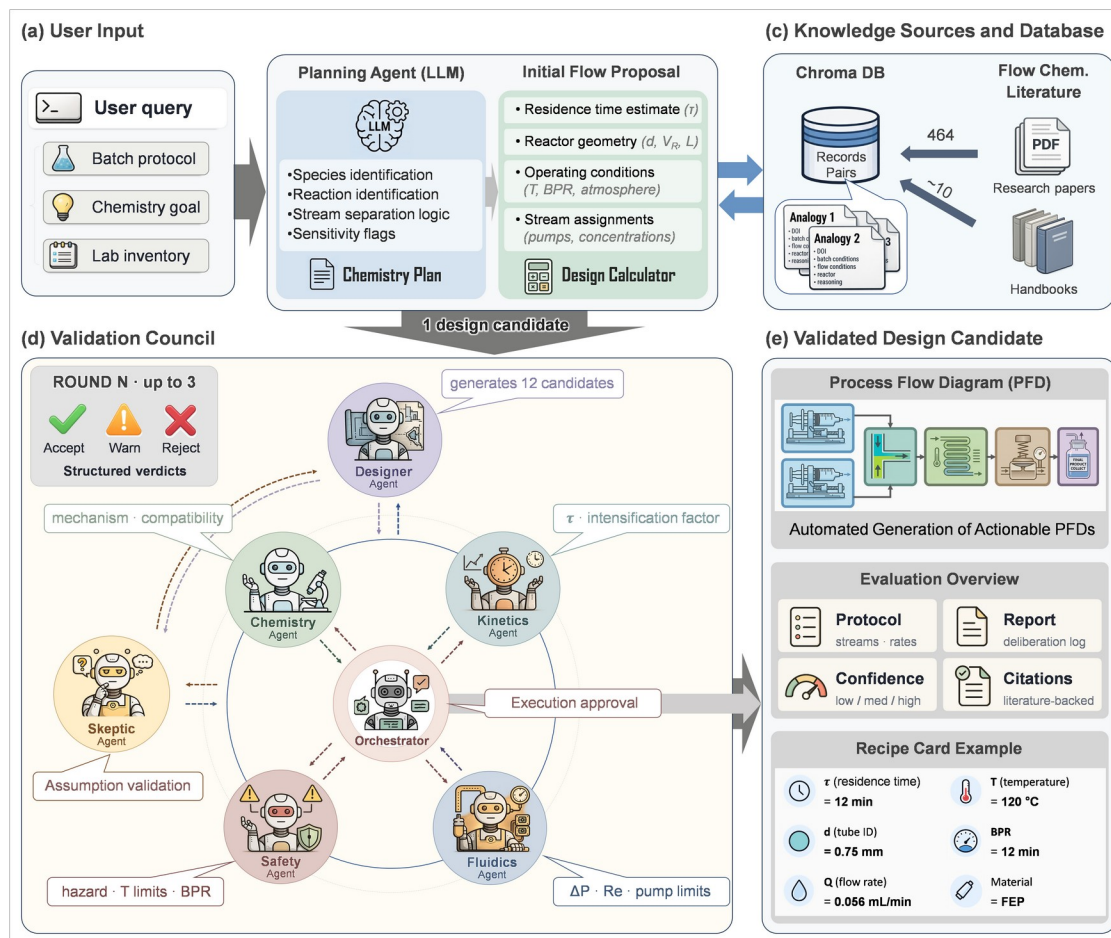


Figure 1. FlowPilot system architecture. User inputs are converted into a chemistry plan and an initial engineering proposal, supported by flow-chemistry literature and handbook knowledge. The seven-agent council generates and reviews candidate designs, records structured verdicts, and proposes bounded revisions. The selected candidate is reconciled with engineering calculations and laboratory inventory before a process-flow diagram, protocol, report, confidence assessment, and supporting citations are returned. The illustrated recipe card is schematic; computational validation does not establish experimental performance.

The downstream council provides the main error-control mechanism. Rather than asking one model to produce a single final answer, the system generates a candidate pool and distributes review across domain-specialized agents. The Designer proposes candidate geometries and operating points; Chemistry Agent evaluates chemical compatibility, photochemical constraints, and stream logic; Kinetics Agent evaluates residence time and conversion plausibility; Fluidics Agent evaluates flow regime, pressure drop, mixing, and hardware compatibility; Safety Agent evaluates pressure, thermal, material, and handling

risks; the Skeptic audits assumptions and internal consistency; and the Orchestrator selects the final candidate. This structure produces not only a selected design but also a record of rejected alternatives, revised candidates, disqualified candidates, and the rationale for each decision.

The output of the workflow is a structured design package rather than only a text answer. It contains a process-flow diagram, stage and stream parameters, equipment assignments, confidence and safety flags, supporting analogies, and the review record. The numerical views and diagram are derived from the same final design contract, while unresolved requirements remain separately visible. This exposes both the proposed operating point and how it was selected without treating a model's confidence as experimental validation. Detailed architecture, authority boundaries, intake contracts, and run provenance are provided in Figures S1-S2 and Tables S1-S3.

Literature and rule-base coverage

The literature dataset covers a broad range of flow-chemistry applications (Figure 2). The curated export contains 464 classified literature records spanning thermal synthesis, photoredox catalysis, heterogeneous photocatalysis, cross-coupling, oxidation/reduction, and other reaction classes. Reactor types range from capillary/coil and microreactor/chip systems to photoreactors and packed beds, with fluoropolymer tubing prominent among the reported materials. Explicit inlet-stream counts are available for 154 processes; two- and three-stream configurations account for 91 and 30 of these records, respectively. These annotations supply precedents rather than prescribing a default number of streams. The diversity supports retrieval across photochemical and non-photochemical processes with different materials and operating regimes.



Figure 2. Literature corpus and engineering rule-base coverage. (a-d) Reaction classes, reactor types, reported materials, and inlet-stream configurations in the curated flow-chemistry corpus; explicit inlet-stream counts are available for 154 records. (e) Category and severity distribution within the 2,537-rule engineering knowledge base. (f) Recorded associations between rule categories and chemistry classes. Full categorical exports and field-specific availability are reported in Figures S3-S4 and Tables S4-S5; associations indicate knowledge coverage rather than independent validation of every rule.

The engineering knowledge base contained 2,537 structured rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety (Figure 2e-f). Coverage was not uniform across chemistry classes: heat-transfer rules were densely associated with thermal synthesis and cross-coupling, whereas photochemistry rules concentrated in photoredox and photocycloaddition classes. This structured rule landscape supplies domain context for

evaluating model-generated designs alongside deterministic checks, rather than relying only on model priors. The frozen corpus composition and computed rule-base coverage are reported in Figures S3-S4 and Tables S4-S5.

Plan-aware retrieval improves analogy quality

The retrieval system was evaluated against a semantic-search baseline (Figure 3). A standard RAG pipeline embeds the batch protocol or query summary and ranks records by cosine similarity. FlowPilot enriches the query using the chemistry plan, including reaction class, mechanism, photocatalyst family, solvent, and wavelength. The filtering sequence starts with paired records under mechanism and phase constraints, relaxes filters when fewer than three matches remain, and expands to all records if no paired match remains. A field-aware score then reranks the retrieved candidates. Figure 3a summarizes this workflow, while the precise tier transitions and scoring weights are specified in Figure S5 and Table S6.

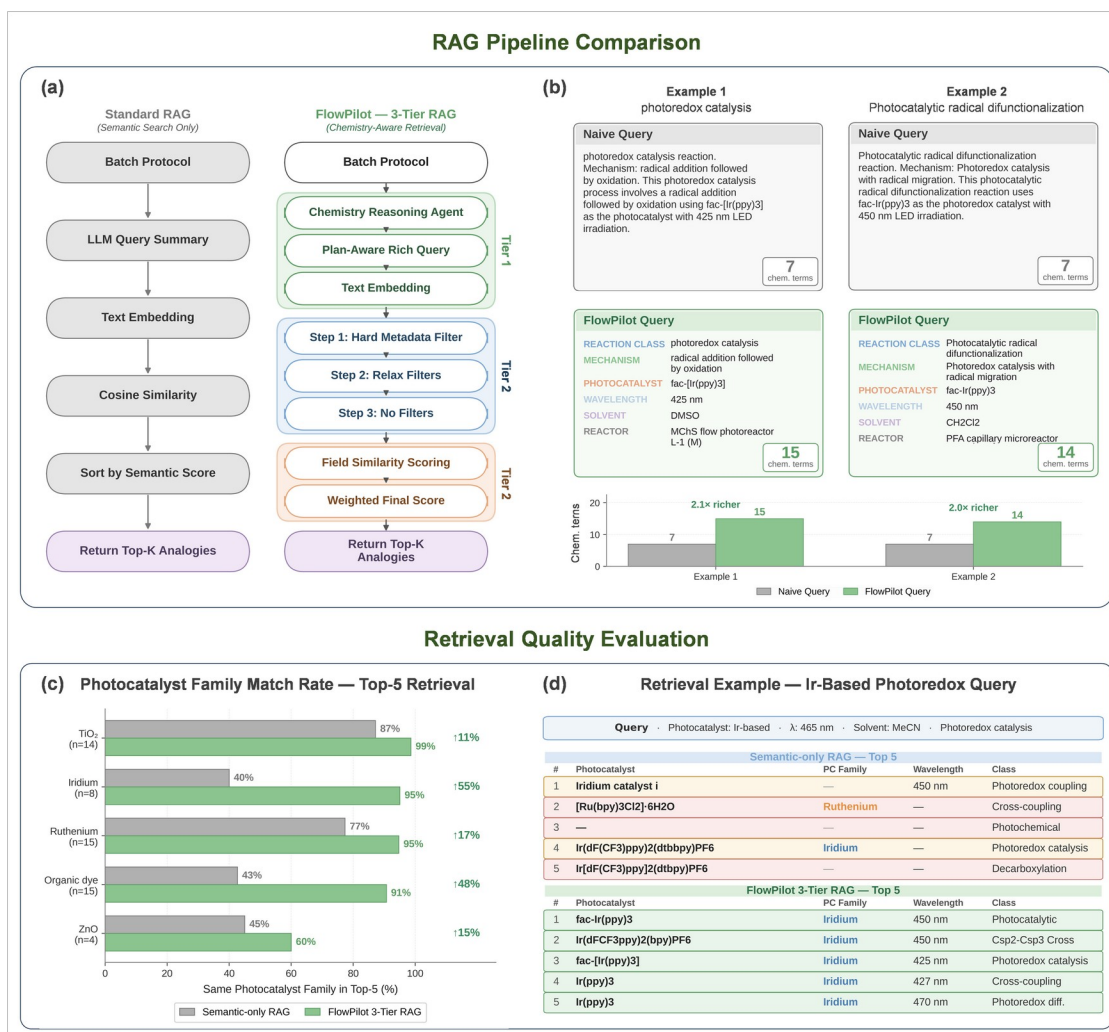


Figure 3. Plan-aware retrieval. (a) Three-tier filtering strategy: hard metadata filter, relaxed metadata filter, and pure semantic fallback. (b) Query enrichment expands chemically specific search terms approximately two-fold relative to a naive semantic baseline. (c) Top-5 photocatalyst-family matching rates by family (iridium, organic dye, ruthenium, TiO₂, ZnO) for semantic baseline vs FlowPilot reranking. (d) Representative reranking outcome showing closer wavelength and photocatalyst-family alignment.

The enriched query representation approximately doubled the number of chemically specific search terms in the two examples shown in Figure 3b. A photoredox-catalysis query increased from 7 naive terms to 15 FlowPilot terms, and a photocatalytic radical difunctionalization query increased from 7 to 14 terms. In the retrieval-pair benchmark, 1,600 query–result pairs were evaluated. FlowPilot reranked 25.1% of these

1,600 pairs relative to the pure semantic baseline, indicating that field-aware scoring materially changed which analogies were presented to the design engine.

The strongest retrieval improvement was observed for photocatalyst-family matching. For iridium systems, top-five family matching increased from 40.0% under semantic search to 95.0% after reranking. Organic-dye systems improved from 42.7% to 90.7%, ruthenium systems from 77.3% to 94.7%, TiO₂ systems from 87.1% to 98.6%, and ZnO systems from 45.0% to 60.0% (Figure 3c). The demonstration in Figure 3d illustrates how field-aware ranking promotes records with a matching photocatalyst family and relevant wavelength context. These results measure metadata alignment, not the accuracy of a transferred kinetic estimate or the eventual reaction yield. Retrieval tiers, scoring weights, benchmark data, and source-exclusion controls are detailed in Figures S5-S6 and Tables S6-S7.

Ablation and architecture comparison

FlowPilot was evaluated through matched architecture comparisons that held the chemistry, inventory, and requested output contract constant while changing whether the generator operated as a one-shot model or within the full pipeline (Figure 4a). The three cases were CuAAC, photochemical oxidation, and hydrogenolysis, with three independent generation repeats per model-architecture-case combination. Independent Qwen, OpenAI, and Claude judge families rated the normalized final designs against 14 fixed criteria, using the same source context and deterministic verification evidence. Applicable criteria received equally weighted integer ratings from 0 to 4, which were averaged across judges and normalized to a benchmark score from 0 to 1. Critical flags were recorded separately from the score. Scoring anchors, the criterion definitions, and module-screen summaries are provided in Tables S8-S10.

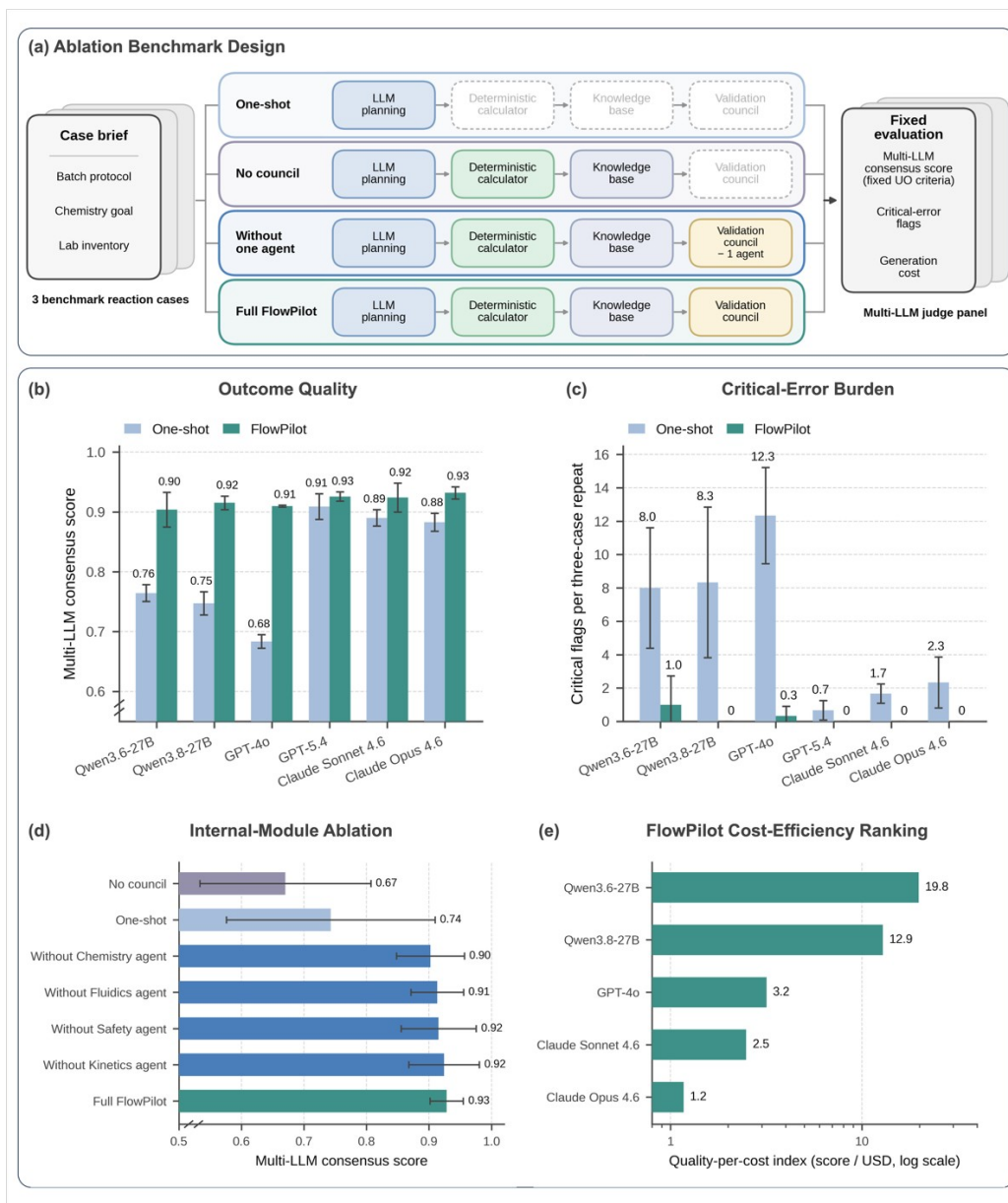


Figure 4. Ablation benchmarking and cost efficiency. (a) Architecture comparison under shared inputs and fixed evaluation. (b) Benchmark scores and (c) raw judge critical flags: mean and sample SD across three three-case repeats. Flags are not independently confirmed physical errors. These panels retain the archived GPT-5.4 comparison, whereas the ESI summary cohort contains five models. (d) Selected Qwen3.8-27B module ablations: mean and between-case SD, with one generation per case-condition. (e) FlowPilot score per generation cost under the recorded price schedule (log scale), excluding deployment costs. Full definitions and case-level results are in ESI Section 3.

Across the five-model cohort in Table S11, full-pipeline mean scores ranged from 0.904 to 0.932, compared with 0.684 to 0.890 for matched one-shot generation (Figure 4b). The mean improvement was largest for GPT-4o (+0.226) and smaller for the Claude models (+0.034 and +0.049). Across 45 outcomes per architecture, the judges recorded 98 critical flags for one-shot and 4 for FlowPilot; 18 and 43 outcomes, respectively, had no critical flag (Figure 4c; Table S12). Thus, the observed advantage concerns consistency and completeness under the specified evaluation, not a universal ranking of models or proof of experimental success. In the internal screen, removing the council produced the largest score decrease among the configurations plotted in Figure 4d, while specialist removals had smaller and chemistry-dependent effects.

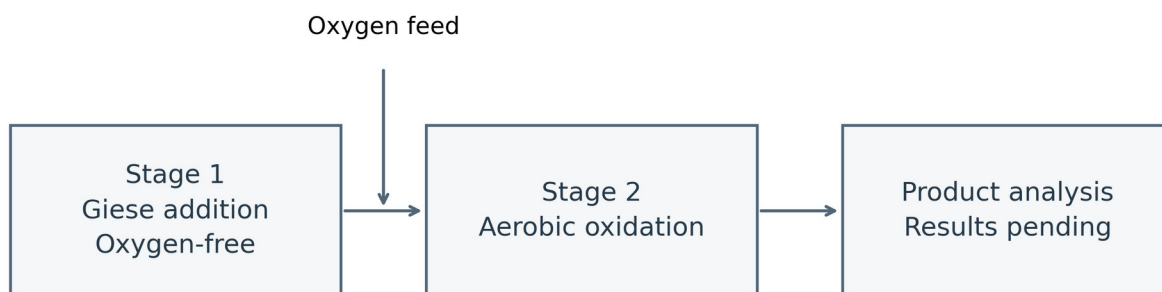
Figure 4e reports the complementary resource trade-off: the Qwen configurations achieved higher score-per-generation-cost values under the recorded token-price schedule. Full-pipeline computation nevertheless includes multiple calls, so user-facing simplicity should not be confused with lower total token use. Figures S7-S10 and Tables S8-S12 provide case-level scores, scoring criteria, module results, critical-flag maps, and resource use. Tables S13-S16 define engineering closure, inventory realization, feedback records, and computational replay. Figure S11 and Table S17 illustrate user-composed input, system-assembled prompts, and returned artifacts for one matched run; the legacy heuristic QA values in that example are distinct from the fixed-criteria multi-LLM score used in Figure 4.

The earlier council model-matrix and candidate-budget experiment is reported separately in Figure S12 and Tables S18-S21. In the five-repeat comparison, the normalized radar-area score changed from 0.31 +/- 0.03 before council review to 0.48 +/- 0.23 afterward. This engineering-profile measure is not the multi-LLM benchmark score in Figure 4. The ESI defines Need revision, New versions, Combined pool, and Disqualified, and reports runtime, token use, geometry, and radar normalization. Figure S13 and Table S22 distinguish the final structured FlowPilot topology from its contemporaneous generated narrative; the complete one-shot response is retained alongside this worked example.

The operational examples connect these evaluations to the chemist-facing workflow. Figures S14-S15 and Table S23 show batch input and fixed-ID follow-up questions; Figure S16 and Table S24 show inventory import, normalization, and remaining equipment warnings. Figure S17 and Table S25 show stage-resolved topology and stream parameters from a reopened run. Table S26 provides concrete one-shot and FlowPilot issues, including an evaluator false positive. Figure S18 and Table S27 retain actual council disagreement and selection records, while Figure S19 and Table S28 place these views in the full GUI and saved-run context. These examples document human interaction and provenance rather than additional wet-lab outcomes.

Case study 1: photoredox Giese addition followed by aerobic oxidation

The first laboratory case couples photoredox Giese addition with aerobic oxidation to a sulfoxide⁴² (Figure 5). The supplied batch sequence uses an oxygen-free first stage followed by oxygen exposure during the second irradiation stage. The revised KHU input therefore requires oxygen introduction only at Stage 2 and equipment selection from the confirmed inventory, without an inline degassing device. This case will assess whether the translated stage configuration, gas delivery, and residence-time choices support the complete reaction sequence. The updated topology and measured outcomes will be added when the experiments are complete.

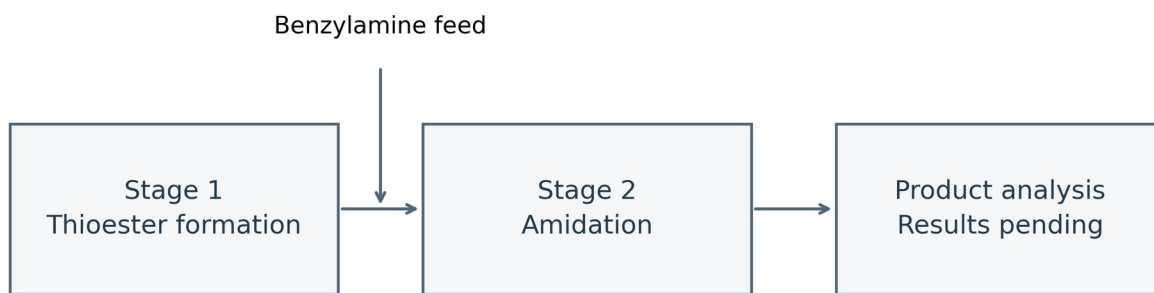


Placeholder: updated topology and wet-lab data to be inserted

Figure 5. Placeholder for the experimental evaluation of photoredox Giese addition followed by aerobic oxidation. The diagram indicates the required oxygen-free first stage and oxygen introduction at Stage 2. The final figure will contain the updated equipment topology, operating conditions, and wet-lab results.

Case study 2: telescoped DPDTC-mediated amidation

The second laboratory case is a two-stage DPDTC-mediated amidation through a thioester intermediate⁴³ (Figure 6). The supplied batch sequence activates 3-methyl-4-nitrobenzoic acid before benzylamine addition, without isolating the intermediate. This case tests interstage feed placement, concentration and molar-flow consistency, and the allocation of reactor volume and residence time across the connected process. The revised KHU inventory and response sets define the available equipment and design objectives. The updated topology, operating conditions, and measured amide yield will be inserted after laboratory evaluation.



Placeholder: updated topology and wet-lab data to be inserted

Figure 6. Placeholder for the experimental evaluation of telescoped DPDTC-mediated amidation. Thioester formation is followed by benzylamine addition and amidation without intermediate isolation. The final figure will contain the updated equipment topology, stream concentrations and flow rates, stage conditions, and wet-lab results.

DISCUSSION

The hybrid architecture separates chemical judgment from engineering arithmetic

The results show how separating chemical interpretation from engineering arithmetic can improve batch-to-flow proposals under a shared evaluation. A model may identify a plausible transformation yet report incompatible stream concentrations, reactor volumes, or residence times. FlowPilot addresses these coupled quantities through calculation and reconciliation, while retaining model-based assessment for chemistry and operating choices. The matched comparisons in Figure 4 test the delivered outcome of this combination, rather than awarding points for the presence of additional agents or longer reasoning traces.

The engineering calculator derives dependent quantities from a common set of inputs, and candidate revisions trigger recalculation before publication. This reduces opportunities for a narrative recommendation to contradict its own volume, flow, and time relationships. Council review adds cross-domain scrutiny, but neither calculation nor deliberation guarantees correct input assumptions. The examples in Tables S22 and S26 make this distinction concrete: a structured result can differ from an accompanying generated narrative, and a judge can misidentify an error. Traceable final parameters and inspectable review records therefore matter as much as a favorable aggregate score.

Positioning relative to prior AI systems for chemistry

Several recent systems have applied LLMs to chemistry tasks: ChemCrow³⁴ augments an LLM with tool calls for property lookup and safety assessment but operates in the batch domain; Coscientist³⁸ demonstrated autonomous experiment execution but targets the laboratory loop rather than process design; El Agente³⁵ automates quantum-chemistry calculations at the electronic-structure layer; Zeng et al.³⁶ applied a multi-agent framework to process optimization but assumes an already-defined process; and Materealize³⁷ employs multi-agent deliberation for materials design but does not incorporate fluid-dynamics or heat-transfer calculations. Classical process simulators²² require a pre-specified flowsheet and do not reason about chemistry, while Bayesian optimizers²⁴⁻²⁸ refine known processes but cannot propose one. FlowPilot addresses the segment between a chemistry brief and a defensible first flow design: it reasons about mechanism, generates engineering calculations, and audits the result through domain-specialist agents, producing a starting point for downstream experimental optimization or rigorous simulation.

The council contribution depends on model and configuration

The matched architecture comparison provides evidence of higher mean scores with FlowPilot for each of the five retained generators, with the size of the improvement depending on the model (Figure 4b). The internal Qwen3.8-27B screen provides a complementary comparison: removing the council lowered the mean score from 0.928 to 0.670 across the three cases (Figure 4d; Table S10). Individual specialist removals produced smaller differences. Because each internal condition-case cell was generated only once, these results support descriptive component analysis rather than a statistically powered claim that every specialist is necessary or that the full configuration is always optimal.

The earlier model-matrix and candidate-budget experiments reveal additional configuration sensitivity (Figure S12). Some model combinations completed only a screening pathway rather than the validated pathway; all Gemma-upstream and GPT-4o-mini-upstream entries in Table S18 are marked screen required. Increasing the candidate budget broadened the explored design families and revision activity without producing monotonically better outcomes (Tables S19-S21). These observations qualify the role of the council: it can revise and compare proposals, but cannot be assumed to rescue every weak upstream result. They also distinguish historical development experiments from the matched model comparison in Figure 4.

Limitations and scope boundaries

Several limitations should be noted. First, computational benchmark performance is not a measurement of reaction yield or laboratory safety. The three benchmark chemistries sample only part of flow-chemistry design space, and three generation repeats provide limited evidence of variability. The laboratory cases in Figures 5 and 6 are intended to test selected proposals against measured outcomes. Integration with experimental platforms⁴⁴ could extend this process, but chemist review and experimental validation remain essential.

Second, gas-flow and residence-time definitions require careful interpretation. Gas equivalents are calculated from inlet molar flow at a declared reference temperature and pressure, not from a gas-to-liquid volumetric ratio. The nominal inlet/STP time, calculated from reactor volume divided by the combined liquid and inlet/STP gas flows, is also not a measured in-channel contact time. Gas compression, dissolution, phase holdup, and slip can separate these quantities.⁴⁵ The ESI retains the distinct time bases used by archived runs (Table S13); comparisons should use a common definition rather than silently relabeling old results.

Third, the engineering rule base (2,537 rules) is not uniformly dense across chemistry classes. Thermal synthesis and photoredox catalysis are well represented, but electrochemistry, biocatalysis, and gas-phase flow chemistry have sparser coverage, which may reduce audit-trail reliability in those domains.

Fourth, the outcome benchmark does not establish performance on long multistep telescopes. Figures 5 and 6 extend the evaluation toward connected two-stage chemistry, where an upstream conversion deficit changes the composition entering the next reactor. Independent stage yields, kinetic information, and operational observations may be needed to interpret final performance and guide subsequent refinement.

Fifth, LLM-based evaluation remains fallible even with fixed criteria and deterministic evidence. Shared model families, incomplete source information, or differences in response presentation may affect judgments. Identity masking is not complete architectural blinding, and multiple critical flags can concern the same underlying defect. The checked false-positive example in Table S26 illustrates why these judgments should remain auditable. Excluding a source paper from retrieval also does not establish that it was absent from model pretraining.

Outlook

The immediate next step is to use the existing feedback interface to connect proposed designs with measured process behavior, preserving the input, equipment profile, actual settings, and observed

response at each iteration. This would support evidence-based refinement and future integration with Bayesian optimization tools^{24–28}. Broader chemistry coverage and longer telescoped sequences are further directions for evaluation. Sustainability assessment would require measured or fully specified material and energy balances: reactor volume, nominal time, and temperature alone do not establish an E-factor or process mass intensity. These extensions build on the same auditable design record rather than assuming that a shorter or smaller process is necessarily better.

CONCLUSION

We have presented FlowPilot as a hybrid multi-agent system for translating a chemistry brief into a traceable continuous-flow process proposal. Standardized intake captures chemist intent and laboratory constraints; chemistry interpretation and literature retrieval inform the design; deterministic calculations reconcile dependent engineering quantities; and a specialist council reviews candidate operating points. Inventory assignment and a shared final design record connect the selected parameters to the process diagram and supporting report.

Matched comparisons across the five-model cohort show higher mean fixed-criteria scores and fewer judge critical flags for FlowPilot than for one-shot generation on the three evaluated chemistries. Module ablations indicate a substantial contribution from council review, while the historical model-matrix and budget studies reveal configuration-dependent limitations. The proposed Giese/oxidation and DPDTC-amidation laboratory evaluations will test whether these computational advantages translate into useful experimental starting conditions.

The practical contribution is to organize the transition from a batch protocol to an inspectable first flow design: the chemist supplies the protocol, answers targeted questions, and binds an inventory, while the system assembles the calculation, review, equipment, and provenance records. This reduces manual information assembly without removing experimental responsibility. FlowPilot is therefore positioned as a design co-pilot whose proposals can be checked, tested, and refined, not as an autonomous guarantee of successful synthesis.

More broadly, the hybrid pattern demonstrated here, LLM-driven interpretation coupled with deterministic computation and multi-agent cross-domain audit, is not specific to flow chemistry. Any engineering design problem that is currently solved by a small team of complementary specialists iterating across coupled domains is a candidate for the same architectural approach. The modular design of FlowPilot is intended to make such extensions practical.

METHODS

A literature corpus containing 464 classified flow-chemistry records was annotated with available reaction, reactor, material, operating-condition, yield, and stream-count metadata. Field completeness was reported separately (Table S4). A complementary handbook-derived engineering knowledge base^{20,21,32} contained 2,537 rules categorized by topic and severity. The rule-base analysis describes coverage and recorded associations, while the deterministic calculator implements the numerical checks used during design.

The plan-aware retrieval pipeline first generates a chemistry plan from the user input and then constructs an expanded query containing reaction class, mechanism, photocatalyst family, and sensitivity flags. Retrieved candidates are reranked by a field-aware score that rewards mechanistic relevance over superficial semantic similarity.

The translation workflow comprises standardized intake, chemistry interpretation, retrieval, engineering calculation, candidate generation, council review, inventory realization, and topology rendering. LLMs propose chemical interpretations and independent design choices; deterministic routines recalculate dependent quantities and check their consistency. Chemist objectives and available evidence guide residence-time and throughput choices. A nominal intensification factor is not treated as proof of conversion, and an inventory-compatible operating point is not itself an experimentally validated procedure. Stage parameters, stream assignments, and topology are reconciled before final publication.

The council reviews a candidate pool under bounded domain authority. Each of the seven agents (Designer, Chemistry Agent, Kinetics Agent, Fluidics Agent, Safety Agent, Skeptic, Orchestrator) may only modify fields within its own domain. Revised candidates are recomputed deterministically before final scoring.

The matched architecture comparison used the same frozen case input and final-output contract for one-shot generation and FlowPilot. Five retained generator models, two architectures, three chemistry cases, and three generation repeats yielded 90 outcomes in the ESI cohort. Three judge families received identity-masked final designs, source context, deterministic verification evidence, and the fixed rubric in Tables S8-S9. Each applicable criterion was rated from 0 to 4; ratings were averaged across judges and criteria, then divided by four. Gas bookkeeping and multistage closure were omitted only when inapplicable. Per-model error bars summarize the sample SD of three repeat-level means, each averaging the three cases. Critical flags were recorded separately. Exact inputs, model records, applicability rules, and archived evaluation limitations are given in ESI Section 3.

The internal module screen fixed the generator to Qwen3.8-27B and generated each condition-case combination once; its SD therefore measures variation between chemistries, not repeatability. Figure 4d shows selected conditions and Table S10 provides the displayed module summaries. The separate historical model-matrix experiment crossed four upstream and four council endpoints, and the candidate-budget study tested $B = 1, 6, 12$, and 24 with five repeats. Their validation dispositions, numerical data, and metric definitions are reported in ESI Section 6.1 and Tables S18-S21. These studies are kept distinct from the independent-judge outcome benchmark.

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Author Contributions

A.M. and G.-N.A. contributed equally to this work. A.M. and G.-N.A. conceived the project and designed the FlowPilot architecture. A.M. developed the upstream chemistry-interpretation agent, the plan-aware retrieval pipeline, and the nine-step deterministic engineering calculator. G.-N.A. constructed the engineering rule base, implemented the Council and orchestration logic, and performed the model-matrix and candidate-budget benchmarks. Both authors designed and analyzed the case studies. A.M. and G.-N.A. authored the manuscript. Both authors read and approved the final manuscript.

Notes

Availability of data and code

Source code and data will be provided to peer reviewers via GitHub during the review process and publicly released upon publication of this work.

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ABBREVIATIONS

BPR, back-pressure regulator; DPDTC, dipyridyldithiocarbonate; Da, Damköhler number; FEP, fluorinated ethylene propylene; IF, intensification factor; LLM, large language model; MFC, mass flow controller; Pe, Péclet number; PFA, perfluoroalkoxy; PFD, process flow diagram; PTFE, polytetrafluoroethylene; RAG, retrieval-augmented generation; Re, Reynolds number; STY, space-time yield.

REFERENCES

- (1) Rossetti, I. Flow Chemistry: New Concepts from Batch to Continuous Organic Chemistry. *Ind. Chem.* **2016**, *2*, e102. <https://doi.org/10.4172/2469-9764.1000e102>.
- (2) Rossetti, I.; Compagnoni, M. Chemical Reaction Engineering, Process Design and Scale-up Issues at the Frontier of Synthesis: Flow Chemistry. *Chem. Eng. J.* **2016**, *296*, 56–70. <https://doi.org/10.1016/j.cej.2016.02.119>.
- (3) Feng, K.; Zhu, Y.-Y.; Gu, S.; Long, J.; Wang, H.-F. Typical Cases of Continuous Flow Chemistry in Pharmaceutical Synthesis in 2023-2024. *Asian J. Org. Chem.* **2025**, *14*, e202500132. <https://doi.org/10.1002/ajoc.202500132>.
- (4) Wegner, J.; Ceylan, S.; Kirschning, A. Flow Chemistry – A Key Enabling Technology for (Multistep) Organic Synthesis. *Adv. Synth. Catal.* **2012**, *354*, 17–57. <https://doi.org/10.1002/adsc.201100584>.
- (5) Cambié, D.; Bottecchia, C.; Straathof, N.; Hessel, V.; Noël, T. Applications of Continuous-Flow Photochemistry in Organic Synthesis, Material Science, and Water Treatment. *Chem. Rev.* **2016**, *116*(17), 10276–10341. <https://doi.org/10.1021/acs.chemrev.5b00707>.
- (6) Padoin, N.; Matiazzo, T.; Riella, H.; Soares, C. A Perspective on the Past, the Present, and the Future of Computational Fluid Dynamics (CFD) in Flow Chemistry. *J. Flow Chem.* **2024**, *14*, 239–256. <https://doi.org/10.1007/s41981-024-00313-4>.
- (7) Movsisyan, M.; Delbeke, E.; Berton, J.; Battilocchio, C.; Ley, S.; Stevens, C. Taming Hazardous Chemistry by Continuous Flow Technology. *Chem. Soc. Rev.* **2016**, *45* (18), 4892–4928. <https://doi.org/10.1039/c5cs00902b>.

- (8) Yasukawa, T.; Kobayashi, S. Translation of Batch to Continuous Flow in Photoredox Reactions. *ACS Central Science*. July 16, 2021, pp 1099–1101. <https://doi.org/10.1021/acscentsci.1c00711>.
- (9) Noël, T.; Cao, Y.; Laudadio, G. The Fundamentals Behind the Use of Flow Reactors in Electrochemistry. *Acc. Chem. Res.* **2019**, *52*, 2858–2869. <https://doi.org/10.1021/acs.accounts.9b00412>.
- (10) Gemoets, H.; Su, Y.; Shang, M.; Hessel, V.; Luque, R.; Noël, T. Liquid Phase Oxidation Chemistry in Continuous-Flow Microreactors. *Chem. Soc. Rev.* **2016**, *45* (1), 83–117. <https://doi.org/10.1039/c5cs00447k>.
- (11) Plouffe, P.; Macchi, A.; Roberge, D. From Batch to Continuous Chemical Synthesis—A Toolbox Approach. *Org. Process Res. Dev.* **2014**, *18*, 1286–1294. <https://doi.org/10.1021/op5001918>.
- (12) Ingham, R. J.; Battilocchio, C.; Fitzpatrick, D. E.; Sliwinski, E.; Hawkins, J. M.; Ley, S. V. A Systems Approach Towards an Intelligent and Self-Controlling Platform for Integrated Continuous Reaction Sequences. *Angew. Chem. Int. Ed.* **2015**, *54* (1), 144–148. <https://doi.org/10.1002/anie.201409356>.
- (13) Adamo, A.; Beingessner, R. L.; Behnam, M.; Chen, J.; Jamison, T. F.; Jensen, K. F.; Monbaliu, J.-C. M.; Myerson, A. S.; Revalor, E. M.; Snead, D. R.; Stelzer, T.; Weeranoppanant, N.; Wong, S. Y.; Zhang, P. On-Demand Continuous-Flow Production of Pharmaceuticals in a Compact, Reconfigurable System. *Science* **2016**, *352* (6281), 61–67. <https://doi.org/10.1126/science.aaf1337>.
- (14) Holtze, C.; Boehling, R. Batch or Flow Chemistry? – A Current Industrial Opinion on Process Selection. *Curr. Opin. Chem. Eng.* **2022**, *36*, 100798. <https://doi.org/10.1016/j.coche.2022.100798>.
- (15) Glasnov, T. N.; Kappe, C. O. The Microwave-to-Flow Paradigm: Translating High-Temperature Batch Microwave Chemistry to Scalable Continuous-Flow Processes. *Chem. Eur. J.* **2011**, *17* (43), 11956–11968. <https://doi.org/10.1002/chem.201102065>.
- (16) Zhang, J.; Wang, K.; Teixeira, A.; Jensen, K.; Luo, G. Design and Scaling Up of Microchemical Systems: A Review. *Annu. Rev. Chem. Biomol. Eng.* **2017**, *8*, 285–305. <https://doi.org/10.1146/annurev-chembioeng-060816-101443>.
- (17) An, H.; Li, A.; Sasmito, A. P.; Kurnia, J. C.; Jangam, S. V.; Mujumdar, A. S. Computational Fluid Dynamics (CFD) Analysis of Micro-Reactor Performance: Effect of Various Configurations. *Chem. Eng. Sci.* **2012**, *75*, 85–95. <https://doi.org/10.1016/j.ces.2012.03.004>.

- (18) Ripken, R. M.; Wood, J. A.; Schlautmann, S.; Günther, A.; Gardeniers, H. J. G. E.; Le Gac, S. Towards Controlled Bubble Nucleation in Microreactors for Enhanced Mass Transport. *React. Chem. Eng.* **2021**, *6*(10), 1869–1877. <https://doi.org/10.1039/D1RE00092F>.
- (19) Mielke, E.; Plouffe, P.; Koushik, N.; Eyholzer, M.; Gottsponer, M.; Kockmann, N.; Macchi, A.; Roberge, D. M. Local and Overall Heat Transfer of Exothermic Reactions in Microreactor Systems. *React. Chem. Eng.* **2017**, *2*(5), 763–775. <https://doi.org/10.1039/C7RE00085E>.
- (20) Capaldo, L.; Wen, Z.; Noël, T. A Field Guide to Flow Chemistry for Synthetic Organic Chemists. *Chem. Sci.* **2023**, *14*(16), 4230–4247. <https://doi.org/10.1039/D3SC00992K>.
- (21) Plutschack, M. B.; Pieber, B.; Gilmore, K.; Seeberger, P. H. The Hitchhiker's Guide to Flow Chemistry. *Chem. Rev.* **2017**, *117*(18), 11796–11893. <https://doi.org/10.1021/acs.chemrev.7b00183>.
- (22) Haydary, J. *Chemical Process Design and Simulation*; Wiley: **2018**. <https://doi.org/10.1002/9781119311478>.
- (23) Reynoso-Donzelli, S.; Ricardez-Sandoval, L. A Reinforcement Learning Approach with Masked Agents for Chemical Process Flowsheet Design. *AIChE J.* **2025**, *71*(1), e18584. <https://doi.org/10.1002/aic.18584>.
- (24) Mottaefegh, A.; Joo, J.-U.; Na, G.-S.; Sharma, V.; Kim, D.-P. AI-Assisted Autonomous Manufacturing of Tailored Drug-Loaded Nanoparticles by Multi-Step Continuous-Flow Platform. *Chem. Eng. J.* **2024**, *500*, 157454. <https://doi.org/10.1016/j.cej.2024.157454>.
- (25) Dunlap, J.; Ethier, J.; Putnam-Neeb, A.; Iyer, S.; Luo, S.-X. L.; Feng, H.; Torres, J. G.; Doyle, A.; Swager, T.; Vaia, R.; Mirau, P.; Crouse, C.; Baldwin, L. Continuous Flow Synthesis of Pyridinium Salts Accelerated by Multi-Objective Bayesian Optimization with Active Learning. *Chem. Sci.* **2023**, *14*, 8061–8069. <https://doi.org/10.1039/d3sc01303k>.
- (26) Ahn, G.-N.; Kang, J.-H.; Lee, H.-J.; Park, B. E.; Kwon, M.; Na, G.-S.; Kim, H.; Seo, D.-H.; Kim, D.-P. Exploring Ultrafast Flow Chemistry by Autonomous Self-Optimizing Platform. *Chem. Eng. J.* **2023**, *453*, 139707. <https://doi.org/10.1016/j.cej.2022.139707>.
- (27) Mottaefegh, A.; Ahn, G.-N.; Kim, D.-P. Meta Optimization Based on Real-Time Benchmarking of Multiple Surrogate Models for Autonomous Flow Synthesis. *Lab. Chip* **2023**, *23*(6), 1613–1621. <https://doi.org/10.1039/D2LC00938B>.

- (28) Jorayev, P.; Soritz, S.; Sung, S.; Jeraal, M.; Russo, D.; Barthelme, A.; Toussaint, F.; Gaunt, M.; Lapkin, A. Machine Learning-Driven Optimization of Continuous-Flow Photoredox Amine Synthesis. *Org. Process Res. Dev.* **2025**, *29*, 1411–1422. <https://doi.org/10.1021/acs.oprd.4c00533>.
- (29) Tu, Z.; Choure, S.; Fong, M. H.; Roh, J.; Levin, I.; Yu, K.; Joung, J.; Morgan, N.; Li, S.-C.; Sun, X.; Lin, H.; Murnin, M.; Liles, J.; Struble, T.; Fortunato, M.; Liu, M.; Green, W.; Jensen, K.; Coley, C. ASKCOS: Open-Source, Data-Driven Synthesis Planning. *Acc. Chem. Res.* **2025**, *58* (11), 1764–1775. <https://doi.org/10.1021/acs.accounts.5c00155>.
- (30) Coley, C.; Green, W.; Jensen, K. Machine Learning in Computer-Aided Synthesis Planning. *Acc. Chem. Res.* **2018**, *51* (5), 1281–1289. <https://doi.org/10.1021/acs.accounts.8b00087>.
- (31) Hartog, P.; Westerlund, A.; Tetko, I.; Genheden, S. Investigations into the Efficiency of Computer-Aided Synthesis Planning. *J. Chem. Inf. Model.* **2025**, *65* (4), 1771–1781. <https://doi.org/10.1021/acs.jcim.4c01821>.
- (32) Guidi, M.; Seeberger, P. H.; Gilmore, K. How to Approach Flow Chemistry. *Chem. Soc. Rev.* **2020**, *49* (24), 8910–8932. <https://doi.org/10.1039/C9CS00832B>.
- (33) Ramos, M. C.; Collison, C. J.; White, A. D. A Review of Large Language Models and Autonomous Agents in Chemistry. arXiv November 14, 2024. <https://doi.org/10.48550/arXiv.2407.01603>.
- (34) M. Bran, A.; Cox, S.; Schilter, O.; Baldassari, C.; White, A. D.; Schwaller, P. Augmenting Large Language Models with Chemistry Tools. *Nat. Mach. Intell.* **2024**, *6* (5), 525–535. <https://doi.org/10.1038/s42256-024-00832-8>.
- (35) Zou, Y.; Cheng, A. H.; Aldossary, A.; Bai, J.; Leong, S. X.; Campos-Gonzalez-Angulo, J. A.; Choi, C.; Ser, C. T.; Tom, G.; Wang, A.; Zhang, Z.; Yakavets, I.; Hao, H.; Crebolder, C.; Bernales, V.; Aspuru-Guzik, A. El Agente: An Autonomous Agent for Quantum Chemistry. *Matter* **2025**, *8* (7), 102263. <https://doi.org/10.1016/j.matt.2025.102263>.
- (36) Zeng, T.; Badrinarayanan, S.; Ock, J.; Lai, C.-K.; Barati Farimani, A. LLM-Guided Chemical Process Optimization with a Multi-Agent Approach. *Mach. Learn. Sci. Technol.* **2025**, *6* (4), 045067. <https://doi.org/10.1088/2632-2153/ae2382>.
- (37) Kim, S.; Choi, J.; Jang, K.; Park, J.; Bernales, V.; Aspuru-Guzik, A.; Jung, Y. Materealize: A Multi-Agent Deliberation System for End-to-End Material Design and Synthesis. arXiv January 22, 2026. <https://doi.org/10.48550/arXiv.2601.15743>.

- (38) Boiko, D. A.; MacKnight, R.; Kline, B.; Gomes, G. Autonomous Chemical Research with Large Language Models. *Nature* **2023**, *624* (7992), 570–578. <https://doi.org/10.1038/s41586-023-06792-0>.
- (39) Guo, X.; Li, Y.; Kong, X.; Jiang, Y.; Zhao, X.; Gong, Z.; Zhang, Y.; Li, D.; Sang, T.; Zhu, B.; Jun, G.; Huang, Y.; Liu, Y.; Xue, Y.; Kundu, R. D.; Lim, Q. J.; Zhao, Y.; Granger, L. A.; Younis, M. B.; Keivan, D.; Sabharwal, N.; Sinha, S.; Agarwal, P.; Vandyck, K.; Mai, H.; Wang, Z.; Venkatesh, A.; Barik, A.; Yang, J.; Yue, C.; He, J.; Wang, L.; Xu, L.; Chen, H.; Wang, J.; Xu, L.; Shetty, R.; Guo, Z.; Song, D.; Jha, M.; Liang, W.; Yan, W.; Zhang, B.; Karnoor, S. B.; Zhang, J.; Pandya, R.; Gong, X.; Ganesh, M. B.; Shi, F.; Xu, R.; Zhang, Y.; Ouyang, Y.; Qin, L.; Rosenbaum, E.; Snyder, C.; Seiler, P.; Dullerud, G.; Zhang, X. S.; Cheng, Z.; Hanumolu, P. K.; Huang, J.; Kulkarni, M.; Namazifar, M.; Zhang, H.; Hu, B. Toward Engineering AGI: Benchmarking the Engineering Design Capabilities of LLMs. arXiv September 21, 2025. <https://doi.org/10.48550/arXiv.2509.16204>.
- (40) Norheim, J. J.; Rebentisch, E.; Xiao, D.; Draeger, L.; Kerbrat, A.; de Weck, O. L. Challenges in Applying Large Language Models to Requirements Engineering Tasks. *Des. Sci.* **2024**, *10*, e16. <https://doi.org/10.1017/dsj.2024.8>.
- (41) Ahn, G.-N.; Shin, J.-A.; Lee, S. Sustainable Continuous Synthesis of 3,5-Disubstituted Isoxazoles: Development of an Integrated Reaction–Separation–Recovery System for Deep Eutectic Solvents. *ACS Sustainable Chem. Eng.* **2025**, *13*, 6589–6597. <https://doi.org/10.1021/acssuschemeng.5c00580>.
- (42) Park, J.; Kim, S. H.; Cho, J.-Y.; Atriardi, S. R.; Kim, J.-Y.; Mardhiyah, H.; Park, B. Y.; Woo, S. K. Batch and Flow Synthesis of Sulfides and Sulfoxides Using Green Solvents and Oxidant through Visible-Light Photocatalysis. *Green Chem.* **2025**, *27*, 3284–3292. <https://doi.org/10.1039/d4gc05769d>.
- (43) Saunders, J. M.; Ocegüera Nava, E.; Li, J.; Wong, M. J.; Freiberg, K. M.; Lipshutz, B. H. Flow-to-Flow Technology: Amide Formation in the Absence of Traditional Coupling Reagents Using DPDTC. *ACS Sustainable Chem. Eng.* **2025**, *13*, 6646–6655. <https://doi.org/10.1021/acssuschemeng.5c00914>.
- (44) Coley, C. W.; Thomas III, D. A.; Lummiss, J. A. M.; Jaworski, J. N.; Breen, C. P.; Schultz, V.; Hart, T.; Fishman, J. S.; Rogers, L.; Gao, H.; Hicklin, R. W.; Plehiers, P. P.; Byington, J.; Piotti, J. S.; Green, W. H.; Hart, A. J.; Jamison, T. F.; Jensen, K. F. A Robotic Platform for Flow Synthesis of

Organic Compounds Informed by AI Planning. *Science* **2019**, *365* (6453), eaax1566.
<https://doi.org/10.1126/science.aax1566>.

- (45) Mallia, C. J.; Baxendale, I. R. The Use of Gases in Flow Synthesis. *Org. Process Res. Dev.* **2016**, *20* (2), 327–360. <https://doi.org/10.1021/acs.oprd.5b00222>.